Making the RV calendar signals robust

The two lookback windows: mean reversion, factor estimation,
and out-of-sample construction

Summary

Our RV calendar screen computes four z-scores per calendar spread and averages them into a single richness score. This note deals with the part of that construction I never had a defensible answer for: the lookback windows.

There are two of them, and they do different jobs.

- W_z, the **z-score lookback**, which sets the reference distribution the score is measured against. Every one of the four signals uses it.
- W_p, the **PCA fit window**, which is the length of panel history the factor basis is estimated on. Only signals 3 and 4 use it.

I had been thinking of these as two instances of the same problem, and they are not. Each one has to be long relative to a completely different quantity, and the two failure modes are independent, so lengthening one cannot compensate for the other being too short. Section 2 lays out the map; Sections 6 and 9 do each one properly.

Four things came out of this that I did not expect when I started:

1. Neither window should be shared across the four signals, and W_z in particular has to differ. The PCA residual reverts several times faster than the raw calendar level, so a single `window` argument applied to all four is necessarily wrong for at least two of them.
2. For signals 3 and 4 we should not be estimating the mean at all. The projection residual is mean-zero by construction, so we can impose $\mu = 0$ and let W_z estimate only the scale. That turns the harder of the two estimation problems into the easier one.
3. Re-reading my own PCA code, the basis is fit on the full panel including the observation being scored, and the whole residual history is reconstructed from that one fit. The consequence I care about is not the look-ahead on today's point, it is that the reference distribution the z-score is measured against was never observable in real time. Our backtest is therefore optimistic and the live series will not reproduce it.
4. Orthogonalising the four signals — which was my first instinct for fixing the window sensitivity — does not address it at all, and if done before the calibration is fixed it makes things worse by propagating the noise from one signal into the others.

A note on the numbers. The parameter values in the worked examples below are illustrative placeholders chosen to show the mechanics. Nothing here has been fitted to our panel yet — that is the first task in Section 13, and these numbers should be replaced with the fitted ones before any of it is quoted.

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1 Where we are now

The current construction, per (expiry, tenor) cell, and which window each signal depends on:

#	Input series being scored	W_p	W_z
1	raw calendar level	—	✓
2	calendar / front-leg price	—	✓
3	PCA residual of the raw calendar panel	✓	✓
4	PCA residual of the normalised panel	✓	✓

Each signal returns $z = (x_T - \hat{\mu})/\hat{\sigma}$ with $\hat{\mu}, \hat{\sigma}$ taken over W_z, and the pipeline averages the four. For signals 3 and 4 the series x being scored is itself the output of a projection estimated over W_p, so those two carry both windows in series.

I see four structural problems, roughly in order of how much I think they cost us.

The reference distribution is assumed Gaussian and stationary

Calendar spreads are skewed, fat-tailed and regime-dependent, so $z = 2$ is not the 97.7th percentile. The part that actually damages us is subtler: $z = 2$ means something *different* for each of the four signals, because they have different tail behaviour. When we average them, whichever signal has the fattest tails produces the largest numbers and quietly dominates the composite. Our weights are not $\frac{1}{4}$ each, and we have no idea what they are.

Both windows encode assumptions we never checked

W_z asserts that the series returns to its mean on a timescale comparable to W_z. W_p asserts that the factor structure is stable over W_p and estimable from it. Neither has been tested. This is the main subject of the note.

The PCA residual is computed in-sample

Section 10.

We do not have four signals

Signals 1 and 2 differ only by the front-leg normalisation; 3 and 4 likewise. My guess is the effective count is near 1.5, and Section 12 gives the statistic to confirm it.

2 Two windows, two failure modes

The reason it helps to separate them is that both windows are estimating a *nuisance object* which then gets subtracted from the price before we score what is left:

- W_z estimates a nuisance **level** — the equilibrium μ — along the time-series direction.
- W_p estimates a nuisance **subspace** — the common factors — along the cross-sectional direction.

In both cases the failure is the same in spirit: if the window is too short, the nuisance estimator is flexible enough to absorb the very dislocation we are trying to detect. A fast-adapting mean chases the level; a fast-adapting factor basis rotates towards the current cross-section. Either way the thing we score has had our signal removed from it.

What differs is the meaning of "too short", and this is the part I had wrong:

	W_z	W_p
Estimates	the level μ	the factor subspace V
Direction	time series	cross section
Must be long relative to	the half-life h	the panel width p
Governing ratio	h/W_z	$q = p/W_p$
Too short ⇒	mean chases the deviation	loadings rotate into the deviation
Too long ⇒	stale equilibrium level	stale factor structure
Quantified by	closed form, eq. (16)	MP threshold, eq. (22)

Because the governing ratios involve different quantities — a persistence timescale on one side, a dimension count on the other — these are genuinely independent constraints. A long W_p does nothing to rescue a W_z that is shorter than the half-life, and vice versa. There is exactly one place they interact, which is an ordering condition derived in Section 9.4.

3 Why an AR(1) is the right model for the W_z side

A z-score presupposes an equilibrium to revert to. The process that makes that precise is Ornstein–Uhlenbeck,

$$dx_t = \kappa(\mu - x_t)\,dt + \sigma\,dW_t, \qquad \kappa > 0. \tag{1}$$

I want to use this rather than something richer for three reasons.

First, it is the simplest process with both a finite stationary variance and a well-defined reversion speed. A random walk has neither, and if a given calendar is behaving like a random walk then no z-score threshold has predictive content in it — which is itself something we should be flagging on the screen.

Second, its two parameters map directly onto what the signal needs: the level to compare against, μ, and the timescale over which that comparison is valid, $1/\kappa$.

Third, its exact discretisation is an AR(1), so estimating it is one OLS regression on data already in the panel. No numerical work and no new data.

3.1 Solution and stationary moments

Integrating (1) from 0 to t,

$$x_t = \mu + (x_0 - \mu)e^{-\kappa t} + \sigma\int_0^t e^{-\kappa(t-s)}\,dW_s. \tag{2}$$

The stochastic integral is Gaussian with zero mean, so

$$\mathbb{E}[x_t \mid x_0] = \mu + (x_0 - \mu)e^{-\kappa t}, \tag{3}$$

$$\mathrm{Var}[x_t \mid x_0] = \sigma^2\int_0^t e^{-2\kappa(t-s)}ds = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t}), \tag{4}$$

and letting $t \to \infty$,

$$x_\infty \sim \mathcal{N}(\mu,\ \sigma_\infty^2), \qquad \sigma_\infty = \frac{\sigma}{\sqrt{2\kappa}}. \tag{5}$$

Equation (5) is what actually justifies the z-score: σ_∞ is the correct scale for a deviation. Worth noting it is not the innovation volatility σ — they differ by $\sqrt{2\kappa}$, so a spread can have large daily moves and a tight stationary distribution, or the reverse.

3.2 Half-life

From (3) a deviation decays by $e^{-\kappa t}$, so the time for half of it to disappear solves $e^{-\kappa h} = \frac{1}{2}$:

$$\boxed{h = \frac{\ln 2}{\kappa}} \tag{6}$$

I think this is the single most useful number we could add to the screen. It sets W_z, the holding period, and the rebalancing frequency at once, and right now we guess all three independently.

4 Estimating the half-life

4.1 Exact discretisation

Sampling (1) at Δt (one business day) gives an AR(1) exactly, not as an approximation:

$$x_t = \mu(1-\phi) + \phi\, x_{t-1} + \varepsilon_t, \qquad \phi = e^{-\kappa \Delta t}, \qquad \mathrm{Var}(\varepsilon_t) = \sigma_\infty^2(1-\phi^2). \tag{7}$$

4.2 Recipe

Regress $x_t = a + b\, x_{t-1} + \varepsilon_t$ over the reference sample and read off

$$\hat\phi = b, \qquad \hat\mu = \frac{a}{1-b}, \qquad \hat\kappa = -\frac{\ln \hat\phi}{\Delta t}, \qquad \boxed{\hat h = -\frac{\ln 2}{\ln \hat\phi}\ \text{days}} \tag{8}$$

with $\Delta t = 1$. Run it separately on each of the four input series and separately per cell. For signals 3 and 4 the series to fit is the out-of-sample residual from Section 10, not the raw calendar — the residual is the thing being scored, so it is the thing whose half-life sets W_z.

4.3 Illustrative magnitudes

To fix ideas, and these are placeholders rather than fits: if the 3m/6m calendar on the 10y gave $\hat\phi = 0.966$ then

$$\hat\kappa = -\ln(0.966) = 0.0346\ \text{day}^{-1}, \qquad \hat h = \frac{0.693}{0.0346} \approx 20\ \text{business days},$$

and if the corresponding residual gave $\hat\phi = 0.87$ we would get $\hat h \approx 5$ days. If the real fits look anything like that then the two series have timescales a factor of four apart and the shared W_z is the first thing to go.

4.4 Finite-sample bias

OLS on (7) is biased downward in ϕ. For an AR(1) with intercept and sample size T,

$$\mathbb{E}[\hat\phi - \phi] \approx -\frac{1+3\phi}{T}. \tag{9}$$

Since h decreases in ϕ we will *under*estimate the half-life, and the effect is amplified for persistent series because (8) is strongly convex near $\phi = 1$. At $T = 500$ and $\phi = 0.97$ the bias in ϕ is about -0.008, which moves the half-life from 23 days to 20: a 13% error in h from a 0.8% error in ϕ. Apply $\hat{\phi}_c = \hat{\phi} + (1 + 3\hat{\phi})/T$ and treat the output as indicative.

4.5 How much we can trust it

By the delta method, with $h(\phi) = -\ln 2 / \ln \phi$,

$$\frac{\partial h}{\partial \phi} = \frac{\ln 2}{\phi (\ln \phi)^2}, \qquad \mathrm{sd}(\hat{h}) \approx \frac{\ln 2}{\hat{\phi}(\ln \hat{\phi})^2}\, \mathrm{sd}(\hat{\phi}). \tag{10}$$

That derivative diverges as $\phi \to 1$, so for persistent cells the interval is wide and badly asymmetric, and I would not trust the delta method there. Better to bootstrap: stationary block bootstrap with blocks of roughly $2\hat{h}$, refit, take empirical quantiles. A cell coming back with a 95% interval of 10 to 80 days is a useful finding — it says W_z for that cell should be ensembled (Section 11) rather than pinned.

5 Checking there is reversion at all

Writing the same regression in differences,

$$\Delta x_t = a + b^* x_{t-1} + \varepsilon_t, \qquad b^* = \phi - 1 = -\kappa \Delta t, \tag{11}$$

testing $b^* = 0$ against $b^* < 0$ is the Dickey–Fuller test. The regression that gives us the half-life also tells us whether the reversion is statistically there, at no extra cost.

The trap I nearly fell into: under the unit-root null the t-statistic on b^* is not t-distributed and not asymptotically normal, so the usual critical values do not apply. For the constant-without-trend case and large T:

Level	10%	5%	1%
Critical value	−2.57	−2.86	−3.43

A t-statistic of -2.2 looks comfortably significant against a normal table and is not significant here. My proposal is that cells failing this get suppressed on the screen rather than shown as a small z-score, because a small z-score reads as "fairly valued" when the honest statement is "this spread has no equilibrium we can measure".

Two details. Use the augmented version, adding lags of Δx, if the residuals are autocorrelated, with lag count by AIC. And because our calendar is a fixed-weight difference of two prices rather than an estimated cointegrating combination, plain Dickey–Fuller critical values are the right ones — if we ever move to estimated hedge ratios we would need Engle–Granger values, which are materially more stringent.

6 The W_z window: how much signal it costs us

We can compute in closed form how much of the true deviation survives estimating the mean over a finite window, so this half of the problem has an analytic answer rather than a backtested one.

6.1 Setup

I will work with an exponentially weighted mean rather than a rectangular window — same idea, better properties, and I argue for the switch on independent grounds in Section 7. Let x_t be stationary OU with reversion rate κ about a constant μ, and let the EWMA estimating the mean have its own decay rate κ_m:

$$\hat{\mu}_t = \kappa_m \int_{-\infty}^{t} e^{-\kappa_m (t-s)} x_s \, ds. \tag{12}$$

What the signal uses is the measured deviation $d_t = x_t - \hat{\mu}_t$, not the true deviation $x_t - \mu$. The question is how far apart they are.

6.2 Derivation

The stationary autocovariance of OU is $\gamma(u) = \mathrm{Cov}(x_t, x_{t-u}) = \sigma_\infty^2 e^{-\kappa|u|}$.

Covariance of the level with its own EWMA:

$$\mathrm{Cov}(x_t, \hat{\mu}_t) = \kappa_m \int_0^\infty e^{-\kappa_m u} \gamma(u) \, du = \kappa_m \sigma_\infty^2 \int_0^\infty e^{-(\kappa_m + \kappa)u} du = \sigma_\infty^2 \, \frac{\kappa_m}{\kappa_m + \kappa}. \tag{13}$$

Variance of the EWMA, with filter $g(u) = \kappa_m e^{-\kappa_m u}$:

$$\begin{aligned}
\mathrm{Var}(\hat{\mu}_t) &= \sigma_\infty^2 \int_0^\infty \int_0^\infty g(u) g(v) \, e^{-\kappa|u-v|} \, du \, dv \\
&= 2\sigma_\infty^2 \kappa_m^2 \int_0^\infty e^{-(\kappa_m+\kappa)u} \left[\int_0^u e^{(\kappa-\kappa_m)v} dv \right] du \\
&= \frac{2\sigma_\infty^2 \kappa_m^2}{\kappa - \kappa_m} \int_0^\infty \left[e^{-2\kappa_m u} - e^{-(\kappa_m+\kappa)u} \right] du \\
&= \frac{2\sigma_\infty^2 \kappa_m^2}{\kappa - \kappa_m} \left[\frac{1}{2\kappa_m} - \frac{1}{\kappa_m + \kappa} \right] = \sigma_\infty^2 \, \frac{\kappa_m}{\kappa_m + \kappa}.
\end{aligned} \tag{14}$$

Conveniently (13) and (14) come out equal, so

$$\mathrm{Var}(d_t) = \mathrm{Var}(x_t) - 2\,\mathrm{Cov}(x_t, \hat{\mu}_t) + \mathrm{Var}(\hat{\mu}_t) = \sigma_\infty^2 \left[1 - \frac{\kappa_m}{\kappa_m + \kappa} \right] = \sigma_\infty^2 \, \frac{\kappa}{\kappa + \kappa_m}. \tag{15}$$

6.3 The retention ratio

Write $h = \ln 2/\kappa$ for the signal's half-life and $H_m = \ln 2/\kappa_m$ for the EWMA's, so $\kappa_m/\kappa = h/H_m$. Define the retention ratio as the fraction of true deviation variance that survives:

$$\boxed{R = \frac{\mathrm{Var}(x_t - \hat{\mu}_t)}{\mathrm{Var}(x_t - \mu)} = \frac{\kappa}{\kappa + \kappa_m} = \frac{1}{1 + h/H_m}} \tag{16}$$

The most useful version is in correlation terms. Since $\mathrm{Cov}(d_t, x_t - \mu) = \mathrm{Var}(d_t)$,

$$\mathrm{Corr}(d_t, \; x_t - \mu) = \frac{\mathrm{Var}(d_t)}{\mathrm{sd}(d_t)\, \sigma_\infty} = \frac{\mathrm{sd}(d_t)}{\sigma_\infty} = \sqrt{R}. \tag{17}$$

So $\sqrt{R}$ is the correlation between the z-score we compute and the z-score we wanted. Information ratio scales roughly linearly with correlation to the true signal, which makes $\sqrt{R}$ an IR haircut directly.

H_m/h	1	2	3	5	10	20
R (variance retained)	0.500	0.667	0.750	0.833	0.909	0.952
$\sqrt{R}$ (IR retained)	0.707	0.816	0.866	0.913	0.953	0.976

If the reference mean adapts as fast as the spread reverts, $H_m = h$, we throw away 29% of the information ratio. The mechanism is intuitive once you see it: the EWMA chases the deviation and absorbs precisely the slow, persistent component, which is the component that was predicting the return. At $H_m = 3h$ the loss is 13%, and by $H_m = 10h$ it is under 5% with essentially nothing left to gain.

6.4 Which way the risk runs

R increases monotonically in H_m, so on this metric longer is always better. That makes the problem asymmetric, which I had not appreciated before doing the algebra:

- The lower bound is hard and quantitative: $H_m \geq 3h$. Below that we destroy signal for nothing.
- The upper bound does not come from (16) at all, because that derivation assumes μ constant. It is not — the equilibrium calendar level drifts across regimes, and a very long H_m compares today against a level that no longer applies. This bound cannot be written in closed form since it depends on how fast the regime moves.

So W_z should sit in $[3h,\ 10h]$, with the choice inside that range ensembled rather than optimised (Section 11). On a 20-day half-life that is 60 to 200 days.

6.5 Grading the W_z we already use

A rectangular window of length W and an EWMA with smoothing α have the same mean lag when $\alpha = 2/(W+1)$, and the EWMA half-life is $H_m = -\ln 2 / \ln(1-\alpha)$. For $W \gg 1$,

$$H_m \approx 0.347\, W. \tag{18}$$

Our $W_z = 252$ is therefore an 87-day half-life. Against a 20-day series half-life that is $H_m/h = 4.4$ and $\sqrt{R} \approx 0.90$, which is fine. Against a 60-day half-life it is 1.45 and $\sqrt{R} \approx 0.77$: we would give up close to a quarter of the IR with nothing on the screen indicating anything was wrong. Which case we are in is empirical and I have not answered it yet, which is why the half-life fits come first.

7 Replacing the rectangular W_z with an EWMA

Separately from the calibration argument, the rectangular window has a specific defect. Our reference set is `series.iloc[-window:]`, so with $W_z = 252$ today's z-score can move because an observation dropped out of the window 252 days ago. A large old value ages out, $\hat{\mu}$ and $\hat{\sigma}$ jump, and the signal changes with nothing having happened in the market. On a mean-reverting strategy that manufactures entries and exits, and it puts a spurious 252-day periodicity into the score's own autocorrelation, which will contaminate any half-life estimated from the score series itself.

The EWMA versions,

$$\hat{\mu}_t = (1-\lambda)\hat{\mu}_{t-1} + \lambda x_t, \qquad \hat{v}_t = (1-\lambda)\hat{v}_{t-1} + \lambda(x_t - \hat{\mu}_{t-1})^2, \qquad \lambda = 1 - 2^{-1/H_m}, \tag{19}$$

have one interpretable parameter, no cliff edge, and degrade gracefully. Use $\hat{\sigma}_t = \sqrt{\hat{v}_t}$, and note the $\hat{\mu}_{t-1}$ in the variance recursion — using $\hat{\mu}_t$ there introduces a small downward bias.

8 Fixing the score: rank-normal instead of z

Even with W_z right, $(x-\hat{\mu})/\hat{\sigma}$ still assumes Gaussianity. Replace it with a rank-based transform. With r_t the rank of x_t among the n observations in the reference set and Φ^{-1} the inverse standard normal CDF,

$$\boxed{z_t^{\mathrm{rn}} = \Phi^{-1}\left(\frac{r_t - 0.5}{n}\right)} \tag{20}$$

the Van der Waerden normal-scores transform. What it buys us:

- Invariance to any monotone transformation of x, so signals 1 and 2 become directly comparable regardless of the front-leg normalisation.
- Immunity to outliers. A single extreme print can no longer distort $\hat{\sigma}$ for the following year.
- All four signals on an identical scale, which is what makes averaging them legitimate. This is the fix for the hidden-weighting problem in Section 1 and I think it is the highest-value line of code in this note.
- Bounded output, so no single signal can dominate the composite with a 6σ reading.

The cost is losing information about the magnitude of genuinely extreme dislocations, which for a richness screen I am willing to pay. If we want it back, a winsorised z-score is the middle ground.

9 The W_p window: estimating the factor subspace

Now the other window. W_p is the number of rows of the calendar panel the PCA is fit on, and it controls something quite different from W_z: not a reference distribution, but *what counts as a common factor*, and therefore what is left behind in the residual.

Let p be the panel width — the number of consecutive-expiry calendars for the tenor — and k the number of components retained. Define the aspect ratio

$$q = \frac{p}{W_p}\,. \tag{21}$$

9.1 Why q is the governing quantity

The failure mode at short W_p is the cross-sectional analogue of a fast-adapting mean. The eigenvectors are estimated from the sample covariance, and with few observations relative to the number of columns they rotate to fit whatever the recent cross-section looks like — including the dislocation we are trying to measure. In the limit $W_p \to p$ the factor space fits the panel almost exactly and the residual goes to zero. We would have built a screen that reports every spread as fairly valued, all the time.

The relevant theory is Marchenko–Pastur. For a p-dimensional panel of W_p independent standardised observations with *no* factor structure at all, the sample correlation eigenvalues do not collapse to one; they spread out to fill

$$\boxed{\left[(1-\sqrt{q})^2,\ (1+\sqrt{q})^2\right]} \tag{22}$$

So any sample eigenvalue below $(1+\sqrt{q})^2$ is indistinguishable from noise, and a "factor" we extract there is a fitted artifact of the estimation window. This gives a principled rule for k rather than a guess, and it couples k to W_p: a shorter window raises the bulk edge and therefore admits fewer genuine factors.

With $p = 8$ calendars:

W_p (days)	60	250	500	1000
$q = p/W_p$	0.133	0.032	0.016	0.008
bulk edge $(1+\sqrt{q})^2$	1.86	1.39	1.27	1.19
min. variance share for a real factor	23%	17%	16%	15%

The last row is the bulk edge divided by $\operatorname{tr} C = p$. At $W_p = 60$ a component must explain 23% of panel variance before we are entitled to call it a factor; at $W_p = 500$, 16%. PC1 on a calendar panel will clear either comfortably. The question this actually settles is whether PC2 and PC3 do, and at $W_p = 60$ I would expect PC2 to be borderline — which means our current $k = 2$ may be extracting one real factor and one noise direction, and subtracting that noise direction from the panel is subtracting part of our signal.

9.2 The diagnostic I would rather rely on

The threshold above is asymptotic and assumes independence, so I do not want to lean on it too hard. The direct empirical check is **subspace stability**: refit the basis at dates s and $s+\Delta$ and measure the principal angles between V_s and $V_{s+\Delta}$, i.e. the singular values $\cos\theta_i$ of $V_s^\top V_{s+\Delta}$. Report

$$\theta_{\max}(\Delta) = \arccos\big(\sigma_{\min}(V_s^\top V_{s+\Delta})\big), \tag{23}$$

the largest principal angle, as a function of Δ and of W_p. If the basis rotates by a material angle over the span of a few weeks, W_p is too short and the residual series is not measuring a consistent quantity through time. This is cheap to compute, it needs no distributional assumption, and unlike the MP bound it tells us about the actual panel.

9.3 Upper bound on W_p

Symmetrically to W_z, a very long W_p gives a stale factor structure. The calendar term structure genuinely changed shape between the ZLB years and the hiking cycle, and a basis averaged across both describes neither. The residual then picks up regime change in the loadings rather than relative value, which will show up as a slow drift in the residual rather than mean-reverting dislocation — and that is detectable, because the Dickey–Fuller test of Section 5 run on the residual series will fail. That is a useful consistency check: **if the residual does not pass a stationarity test, W_p is probably too long**, independently of anything to do with W_z.

9.4 The ordering condition

The one place the two windows interact. The reference distribution for signals 3 and 4 is the set of past residuals $\{e_s\}$ over the span of W_z. For their dispersion $\hat{\sigma}$ to mean anything, those residuals

must all be measuring the same quantity — which requires the basis to be approximately fixed across that span. If the basis rotates materially within W_z, then $\hat{\sigma}$ is a mixture of genuine residual variation and basis rotation, and today's residual is being compared against residuals defined in a different coordinate system.

So the condition is not really "$W_p > W_z$" as a comparison of two numbers, it is:

$$\boxed{\theta_{\max}(W_z) \text{ must be small}} \tag{24}$$

the basis rotation timescale must exceed W_z. In practice that does imply $W_p \gg W_z$, since the rotation timescale is set by W_p, but stating it as a rotation condition is what makes it checkable rather than a rule of thumb. It also makes the direction of the constraint clear: it is W_z that has to be short enough for the basis to be stable over it, not W_p that has to be long for its own sake.

10 Signals 3 and 4: taking the PCA out of sample

10.1 What our implementation does

In `signal_3_pca_residual_zscore` the basis is fit on the full panel, including the final row, and the entire residual history is reconstructed from that single fit. Two things follow.

Today's observation helped define the factors it is measured against, which shrinks its own residual. The influence of one row falls with W_p, so on a long panel I am not very worried about this.

The second matters more. The historical residuals we build $\hat{\sigma}$ from were never observable in real time. In-sample reconstruction residuals are systematically smaller than out-of-sample ones — and by (22) the gap widens as q grows — so $\hat{\sigma}$ is too small and every z-score is inflated. Unlike the first issue this does not go away with more data. The backtest overstates and the live signal will not match it.

10.2 The construction I propose

Build the residual series recursively. For each date s:

1. Fit on strictly prior data only, the W_p rows ending at $s-1$: column mean m_s and loadings $V_s \in \mathbb{R}^{p\times k}$ from the top k eigenvectors.

2. Project the current row onto that frozen basis:

$$\hat{x}_s = m_s + V_s V_s^\top (x_s - m_s), \qquad e_s = x_s - \hat{x}_s = (I - V_s V_s^\top)(x_s - m_s). \tag{25}$$

3. Take the target column $e_s^{(j)}$.

Then score $e_T^{(j)}$ against $\{e_s^{(j)}\}_{s<T}$ over W_z. The refit need not be daily — weekly or monthly with a frozen basis in between is fine, closer to how we think about the factors anyway, and it makes the rotation condition of Section 9.4 easier to satisfy.

10.3 And do not estimate the residual's mean

Equation (25) shows e_s is the projection of a demeaned vector onto the orthogonal complement of the factor space, so its unconditional mean is zero by construction, up to instability in the basis. So

$$z_t = \frac{e_t^{(j)}}{\hat{\sigma}_t}, \qquad \text{with } \mu \equiv 0 \text{ imposed rather than estimated.} \tag{26}$$

Worth spelling out because it removes the retention problem of Section 6 entirely for signals 3 and 4. The attenuation in (16) comes from an estimated mean chasing the signal; if the mean is known a priori there is nothing to chase. What is left for W_z to do is estimate a scale, and variances converge much faster than means, so a shorter W_z is safe here — which is convenient, because Section 9.4 wants W_z short for these two signals anyway. The two constraints point the same way.

10.4 One thing to check while I am in there

Whether the PCA is fitted on levels or differences. If we fit levels of a non-stationary panel then PC1 is largely trend, the denoising is not doing what it is supposed to, and the MP threshold in (22) does not apply either, since it assumes standardised observations. I want to look at what PC1 actually is before trusting signals 3 and 4 at all.

11 Ensembling the windows instead of choosing

Given how wide the bootstrap interval on $\hat{h}$ is likely to be, pinning a single W_z is false precision. Treat it as a nuisance parameter and average: compute each signal at $H_m \in \{3h, 5h, 10h\}$ and average the scores. The multi-window scores will correlate around 0.7 to 0.9, so this costs almost no signal while removing the window-timing noise and the cliff-edge artifact. The part I value most is procedural: it removes the temptation to re-optimise every time performance dips, which is how this kind of strategy usually dies.

I would not ensemble W_p the same way. Averaging over factor bases is not averaging over a nuisance location — the residuals from different W_p live in different subspaces and averaging them has no clean interpretation. For W_p I would pick one value from the stability diagnostic and the MP threshold, and report the sensitivity of the final signal to it as a robustness check rather than folding it into the score.

11.1 The disagreement is worth reporting

Alongside the composite, show

$$\Delta z = z^{\text{short}} - z^{\text{long}}. \tag{27}$$

If the short-window score is $+2$ and the long-window score is -0.5, the spread is high against recent history but low against the cycle, meaning the recent regime has been unusually depressed. That is a regime flag, free given the ensemble is already computed, and legible on the desk without explanation: cheap versus the last six months, not cheap versus the cycle. It is also the empirical diagnostic for the W_z upper bound in Section 6 — a persistently large $|\Delta z|$ says μ is drifting and the long window has gone stale.

12 Redundancy, and where orthogonalisation belongs

12.1 Measuring it

Let C be the 4×4 correlation matrix of the four scores through history with eigenvalues $\lambda_1, \ldots, \lambda_4$. The effective number of independent signals is the participation ratio

$$\boxed{N_{\text{eff}} = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2} = \frac{16}{\sum_i \lambda_i^2}} \tag{28}$$

using $\sum \lambda_i = \operatorname{tr} C = 4$. Four means genuinely independent, one means a single signal computed four ways. I expect near 1.5, and that number belongs in the strategy documentation rather than being left implicit.

12.2 Why I am not going to use Gram–Schmidt

Sequential orthogonalisation was my first instinct and I now think it is wrong here. It is order-dependent, and there is no economic argument for why signal 1 should be the base rather than signal 3 — different orderings give different composites with no principled way to choose. More importantly it propagates calibration error: if signal 1 is noisy because its W_z is badly set, regressing the others on it injects that noise into all three residuals. Orthogonalising before fixing the calibration actively makes things worse, which is the opposite of what I expected when I started looking at this.

12.3 What to do instead

Shrunk inverse-correlation weights:

$$C_\delta = (1-\delta)C + \delta I, \qquad w \propto C_\delta^{-1}\mathbf{1}, \qquad w \leftarrow \frac{w}{\mathbf{1}^\top w}. \tag{29}$$

The shrinkage is not optional. A 4×4 correlation matrix of series correlating 0.9 is ill-conditioned and inverting it raw produces large offsetting weights that are pure noise amplification — the same q problem as Section 9, in miniature. I would use $\delta \in [0.2, 0.5]$ and check the weights are stable through time.

Being honest about the size of this: with four signals this correlated the gain from clever weighting is second-order. Equal-weighting the two groups — one level score from 1 and 2, one residual score from 3 and 4 — captures most of the benefit with no estimation risk, and that is what I would ship first.

13 How I intend to validate this

13.1 Plateaus, not peaks

For either window, plot IR against it and read the shape rather than the maximum. If IR is 0.9 at 250 days, 0.15 at 125 and 0.2 at 500, that spike is overfitting, because a real effect is smooth in its parameters. If IR runs 0.5 to 0.6 across 125 to 500 then the signal is real, and I take the middle of the plateau and leave it alone. The existence of the plateau is the result; its height is not.

13.2 Purging and embargoing the cross-validation

The holding period is roughly h days, so neighbouring observations have overlapping outcome windows and naive k-fold cross-validation leaks — a training point at t shares its outcome window with a test point at $t+1$. Purge from training any observation whose outcome window overlaps the test block, and embargo a further h days after each block. For signals 3 and 4 the purge also has to respect W_p: a test point must not be scored against a basis fitted on data inside the test block.

13.3 Haircutting for the search

Count the configurations honestly: four signals, six values of W_z, three values of W_p and three of k is a large number. The maximum Sharpe across N trials is inflated even with no edge anywhere — for independent trials the expected maximum of N standard normals grows like $\sqrt{2\ln N}$, about 2.9 at $N = 72$. With $T = 1000$ days the standard error on an annualised Sharpe is around $\sqrt{252/T} \approx 0.5$, so a Sharpe of 1.4 is reachable by selection alone. Our trials are highly correlated so the true haircut is smaller, but it is not zero, and I would rather state the number of configurations tested than have someone ask later.

14 Proposed configuration and sequencing

Signal	W_p	W_z	Score
1 raw calendar	—	$[3h_1, 10h_1]$, ensembled, EWMA	rank-normal
2 normalised calendar	—	$[3h_2, 10h_2]$, ensembled, EWMA	rank-normal
3 PCA residual (raw)	stability + MP	scale only, short, $\mu \equiv 0$	rank-normal
4 PCA residual (norm.)	stability + MP	scale only, short, $\mu \equiv 0$	rank-normal

with each h_i estimated per signal and per cell from (8), bias-corrected by (9), and W_p set from the subspace-stability diagnostic subject to the MP threshold (22) admitting the k we want.

Order of work

1. Out-of-sample PCA projection, Section 10. A defect rather than a tuning choice, it changes the numbers, and it is a few lines.

2. Rank-normal scores, (20). Fixes the implicit weighting in the composite. Also a few lines.

3. Impose $\mu = 0$ on signals 3 and 4, which reduces their W_z to a scale window.

4. Check the eigenvalue spectrum against (22) at our current W_p, and look at whether PC1 is trend. This decides whether $k = 2$ is defensible at all, and it is the cheapest of the diagnostics.

5. Compute $\theta_{\max}(\Delta)$ for a few candidate W_p and fix W_p from it, then verify the rotation condition of Section 9.4 against the W_z we intend to use.

6. Fit half-lives (8) per cell and per signal — on the out-of-sample residual for 3 and 4 — report the Dickey–Fuller statistic alongside, and suppress cells that fail. This is where the placeholder numbers in this note get replaced.

7. Move to EWMA and ensemble W_z, Sections 7 and 11; add Δz to the screen as a regime column.

8. Report N_{eff} (28), then revisit weighting (29).

Items 1 to 4 are cheap and I expect them to move the numbers materially. Item 8 is where the temptation is and where the least money is, so it goes last.